Exam-2 Practice Question

Section A: Multiple Choice

Q1. Why is scaling (dividing by $\sqrt{d_k}$) used in self-attention?
A. Prevent overfitting
B. Stabilize gradients when dot products grow large
C. Reduce model size
D. Improve sparsity

Q2. Which statement about contrastive learning is TRUE?
A. It requires labeled data
B. It always needs negative samples
C. It learns representations via similarity structure
D. It directly optimizes classification accuracy

Q3. Which component is MOST responsible for enabling parallelization in Transformers?
A. Positional encoding
B. Self-attention
C. Feed-forward network
D. Masking

Q4. Why might pretext tasks fail in SSL?
A. Too much data
B. Model too simple
C. Learning shortcuts unrelated to semantics
D. Lack of optimization

Q5. Residual connections in Transformers primarily help:
A. Reduce parameters
B. Prevent vanishing gradients.
C. Improve interpretability
D. Increase sparsity

Q6. Why is evaluating the performance of an outlier detection model inherently difficult?
A. There is a lack of labeled anomalies and severe class imbalance.
B. Anomaly detection models cannot output measurable scores.
C. Outlier detection models always overfit to the majority class.
D. Most evaluation metrics like F1 and ROC-AUC are incompatible with unsupervised learning.

Section B: Short Answer

Q7. Why are activation functions essential in deep neural networks?

Q8. How do reconstruction-based methods (like PCA or Autoencoders) identify outliers?

Q9. Why is contrastive learning generally more robust than pretext-task-based SSL?

Q10. Explain why LSTM is difficult to parallelize compared to Transformers.

Q11. What would happen if masking is removed from the Transformer decoder during training?

Section C: Analytical Problems

Q12. Self-Attention Computation
Given:

- Query vector:
  $q = [1, 0]$
- Keys:
  $k_1 = [1, 0]$, $k_2 = [0, 1]$
- Values:
  $v_1 = [2, 0]$, $v_2 = [0, 3]$

(a) Compute attention scores using dot product
(b) Apply softmax
(c) Compute final output

Q13. Backward Propagation
(See examples from lectures and review)

Section D: Conceptual Comparison

Q14. Compare **RNN/LSTM vs Transformer** in terms of:

- Information flow
- Long-range dependency modeling
- Computational complexity

Q15. Explain the relationship between:

- SSL
- Foundation Models
- Fine-tuning

Why is SSL critical for modern AI systems?

Section E: Open-Ended

Q16 (Open-Ended – Theory & Critique)
Self-supervised learning relies heavily on the assumption that solving a pretext task leads to useful representations.
Question:

- When can this assumption fail?
- Provide at least **two concrete failure scenarios**
- Propose **one improvement or alternative approach**

Q17 (Open-Ended – Design Question)
You are given a **large unlabeled physiological time-series dataset** (e.g., ECG).
Design a **self-supervised learning framework**:

- Define a pretext or contrastive task
- Explain how data augmentation or views are generated
- Describe how you would evaluate the learned representation
- Discuss one potential limitation